

Fórmulas para el primer parcial

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$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad \mathbb{E}(X) = np \quad \mathbb{V}(X) = np(1-p)$$

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$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \mathbb{E}(X) = \lambda \quad \mathbb{V}(X) = \lambda$$

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$$f(x) = p(1-p)^{x-1} \quad \mathbb{E}(X) = \frac{1}{p} \quad \mathbb{V}(X) = \frac{1-p}{p^2}$$

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$$f(x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} \quad \mathbb{E}(X) = n \frac{M}{N} \quad \mathbb{V}(X) = n \frac{M}{N} \left(1 - \frac{M}{N}\right) \frac{N-n}{N-1}$$

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$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{si } x \geq 0 \\ 0 & \text{c.c.} \end{cases} \quad F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{si } x \geq 0 \\ 0 & \text{c.c.} \end{cases} \quad \mathbb{E}(X) = \frac{1}{\lambda} \quad \mathbb{V}(X) = \frac{1}{\lambda^2}$$

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$$f(x) = \begin{cases} \frac{1}{b-a} & \text{si } a \leq x \leq b \\ 0 & \text{c.c.} \end{cases} \quad F(x) = \begin{cases} 0 & \text{si } x < a \\ \frac{x-a}{b-a} & \text{si } a \leq x < b \\ 1 & \text{si } x \geq b \end{cases} \quad \mathbb{E}(X) = \frac{a+b}{2} \quad \mathbb{V}(X) = \frac{(b-a)^2}{12}$$

Derivadas de funciones básicas

- $(e^x)' = e^x$
- $(x^n)' = nx^{n-1} \quad n \neq 0$
- $(\text{Constante})' = 0$
- $(\ln x)' = \frac{1}{x}$

Reglas de derivación

- $(kf(x))' = kf'(x)$ con k constante
- $(f(x) \pm g(x))' = f'(x) \pm g'(x)$
- $(f(x) \cdot g(x))' = f'(x)g(x) + f(x)g'(x)$
- $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$
- $(f(g(x)))' = f'(g(x)) \cdot g'(x)$

Reglas de integración

- $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$
- $\int kf(x) dx = k \int f(x) dx, \quad \text{con } k \text{ constante}$

Integrales de funciones básicas

- $\int_a^b x^n dx = \frac{x^{n+1}}{n+1} \Big|_a^b \quad n \neq -1$
- $\int_a^b \frac{1}{x} dx = \ln(x) \Big|_a^b, \quad \text{con } a, b > 0$